

Communicating with Data – Color

Dr. Ab Mosca (they/them)

Slides based off slides courtesy of Jordan Crouser (<https://jcrouser.github.io/>)

Designing Visual Stories & Graphics

MONDAY
NOV 2
KATHLEEN
MCCARTNEY
HALL

6-7PM
KMH 106

Panel Conversation

Moderated by **Ab Mosca**,
Visiting Assistant Professor
in Computer Science and
Statistical & Data Sciences



Francesca Paris is a reporter for The Upshot, a section of The New York Times specializing in explanatory, analytical, and visual journalism.



Tanvi Sharma is a design technologist working at the intersection of data, aesthetics, and the humanities.



Ally Levine writes, illustrates and develops visually-driven stories on the Reuters Graphics Desk.

7:15-
8:15PM

Interactive Workshops

Draw Your Own Chart

Pick a topic, find data, look at some different chart forms and then use paper, pens, tape, and glue to make physical representations of the numbers.

KMH 101

Visualizing Histories: A Peabody-Inspired Data Workshop

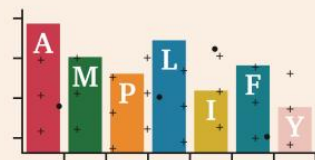
Drawing from your own calendar, select a timeframe — maybe a month — to document events, asking: what do we remember, forget, or want to forget?

KMH 106

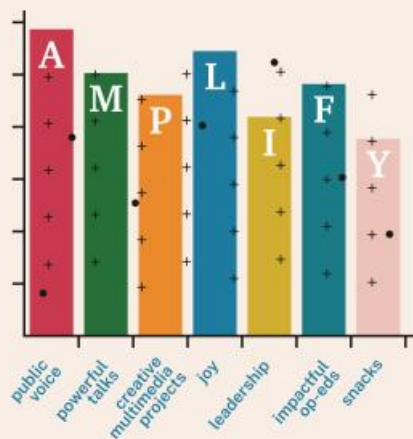
Hey Data, Make Me Feel Something

Go beyond simply showing the trend and explore how to visually represent data in ways that evoke the feeling of the story.

KMH 201



SMITH COLLEGE
Wurtele Center
for Collaborative Leadership



SMITH COLLEGE
Wurtele Center
for Collaborative Leadership

What is Amplify?

Amplify is a program for Smith students to gain the skills, coaching, and platform you need to share your knowledge and perspectives with a public audience.

Events and workshops focus on practices of public voice including public writing, speaking, and other creative mediums.

The program culminates with a chance to participate in the Amplify Competition, which is open to all Smith undergraduates.

Get Inspired

MONDAY 11/2 IN MCCARTNEY HALL

6-7PM: PANEL CONVERSATION

Designing Visual Stories and Graphics

Moderated by **Ab Mosca**, Visiting Assistant Professor in Computer Science and Statistical & Data Sciences



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7:15-8:15PM: WORKSHOPS

Draw Your Own Chart
McCartney Hall 101

*Visualizing Histories:
A Peabody-Inspired Data Workshop*
McCartney Hall 106

Hey Data, Make Me Feel Something
McCartney Hall 201

Create & Submit

EARLY JANUARY

Prepare a submission for Amplify! Write an op-ed, develop a public talk, or design a visual story or graphic

DURING J-TERM

Get some feedback or coaching on your Amplify submission from Wurtele Center staff

WEDNESDAY 1/20

Last day to register for the Public Speaking Competition and submit to the Public Writing and Wild Card categories

SATURDAY 1/30

Registered Public Speakers attend workshop with Emily Brown '11, Public Speaking and Embodied Voice Coach

Share & Celebrate

MONDAY 2/1

Amplify Gallery Website goes LIVE, People's Choice voting opens

SATURDAY 2/13: AMPLIFY DAY!

10AM-NOON

Public Speaking Dress Rehearsal

1-3PM

Public Speaking Competition (plus audience trivia and epic snacks!)

3-5PM

Reception and Gallery of Op-Eds and Visual Stories and Graphics

5-6PM

Award Ceremony

Categories & Prizes

3x
\$500

PUBLIC SPEAKING:

Deliver a 5-minute TED-style talk on Amplify Day

3x
\$500

PUBLIC WRITING:

Submit an op-ed of 750-1,000 words

3x
\$500

WILD CARD:

VISUAL STORIES AND GRAPHICS use imagery, colors, typography, and layout to convey messages, narratives, and data rather than relying solely on text

3x
\$200

PEOPLE'S CHOICE

One award in each category

For More Info



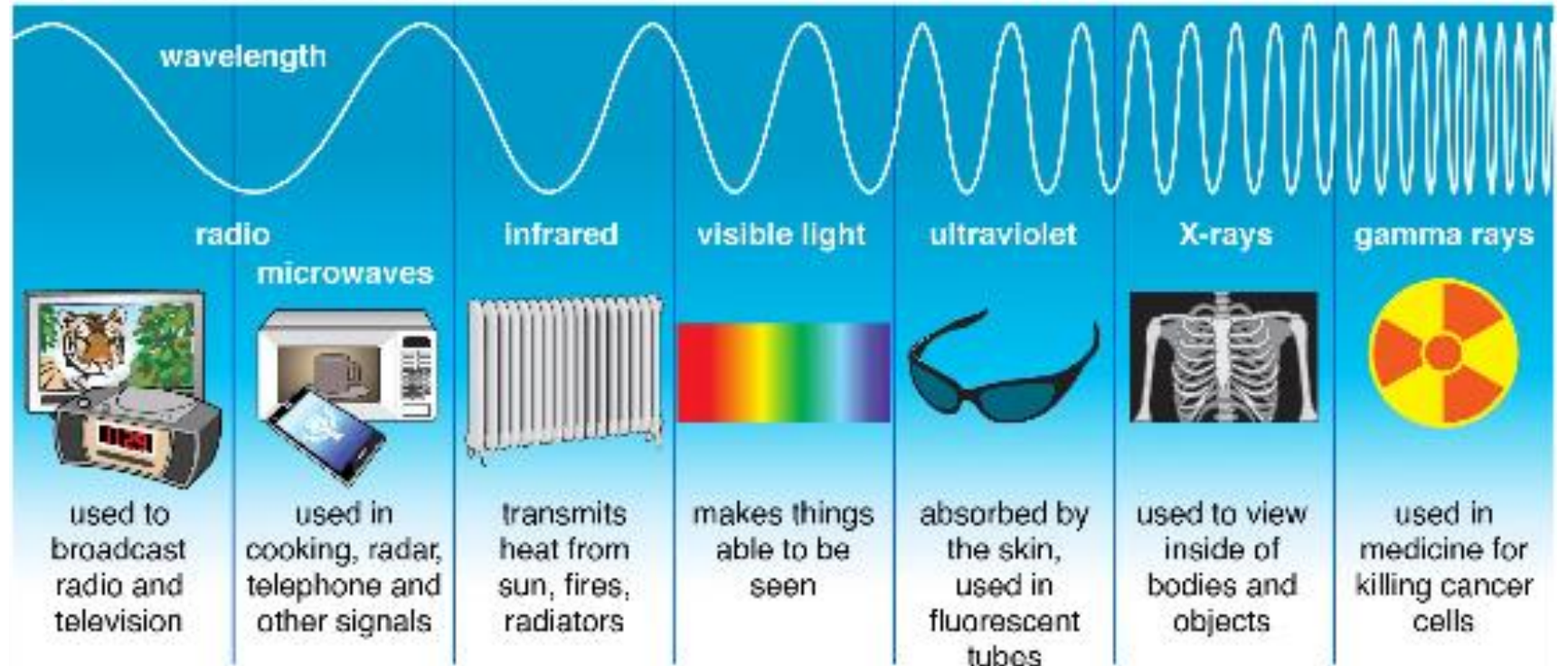
If you have questions, need help getting started, or want to get on our mailing list to get all the updates, reach out to us at wurtelecenter@smith.edu.

Plan for Today

- Electromagnetic radiation
- How we see color
- Color for visual encoding
- More perceptual problems

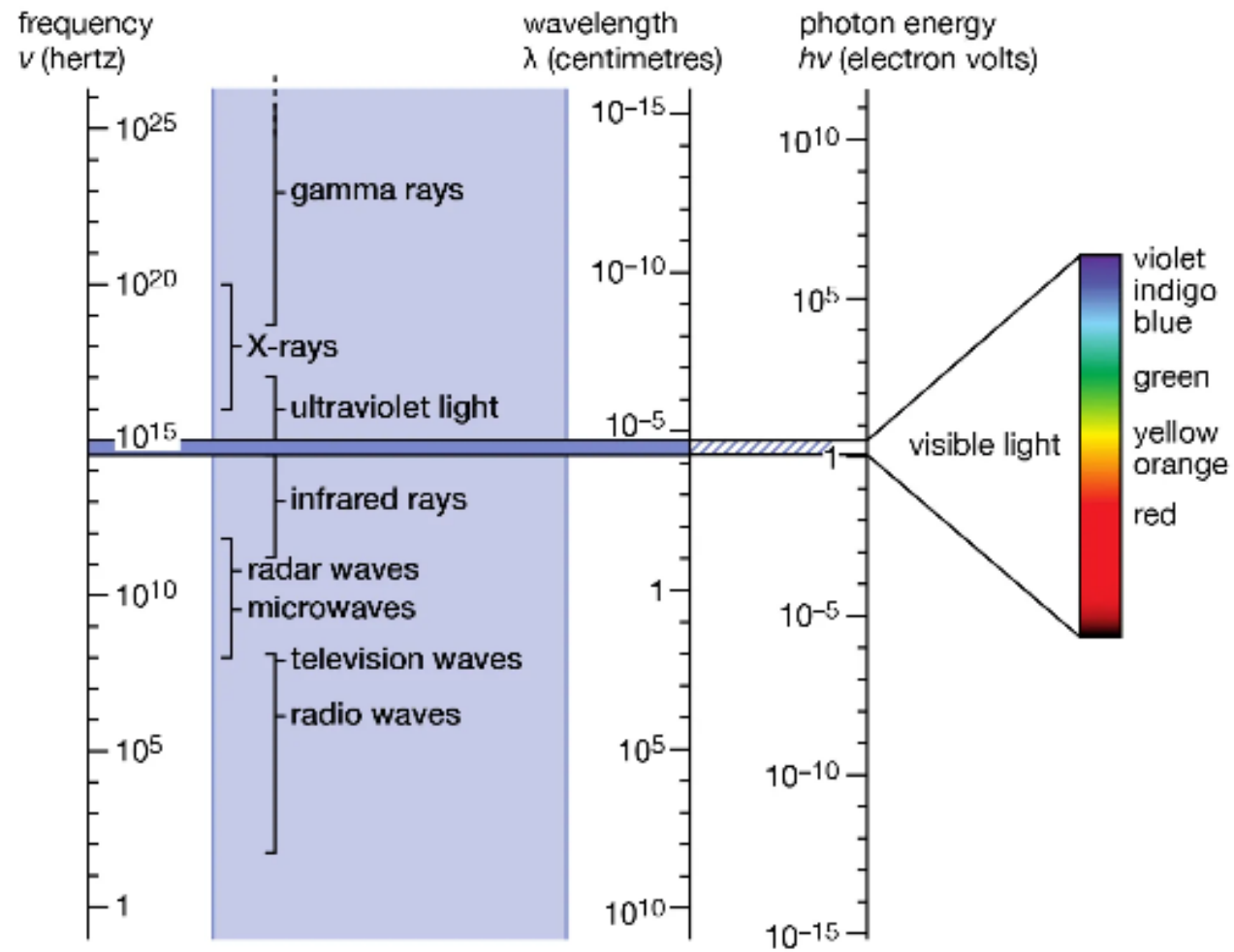
Electromagnetic Radiation

Types of Electromagnetic Radiation



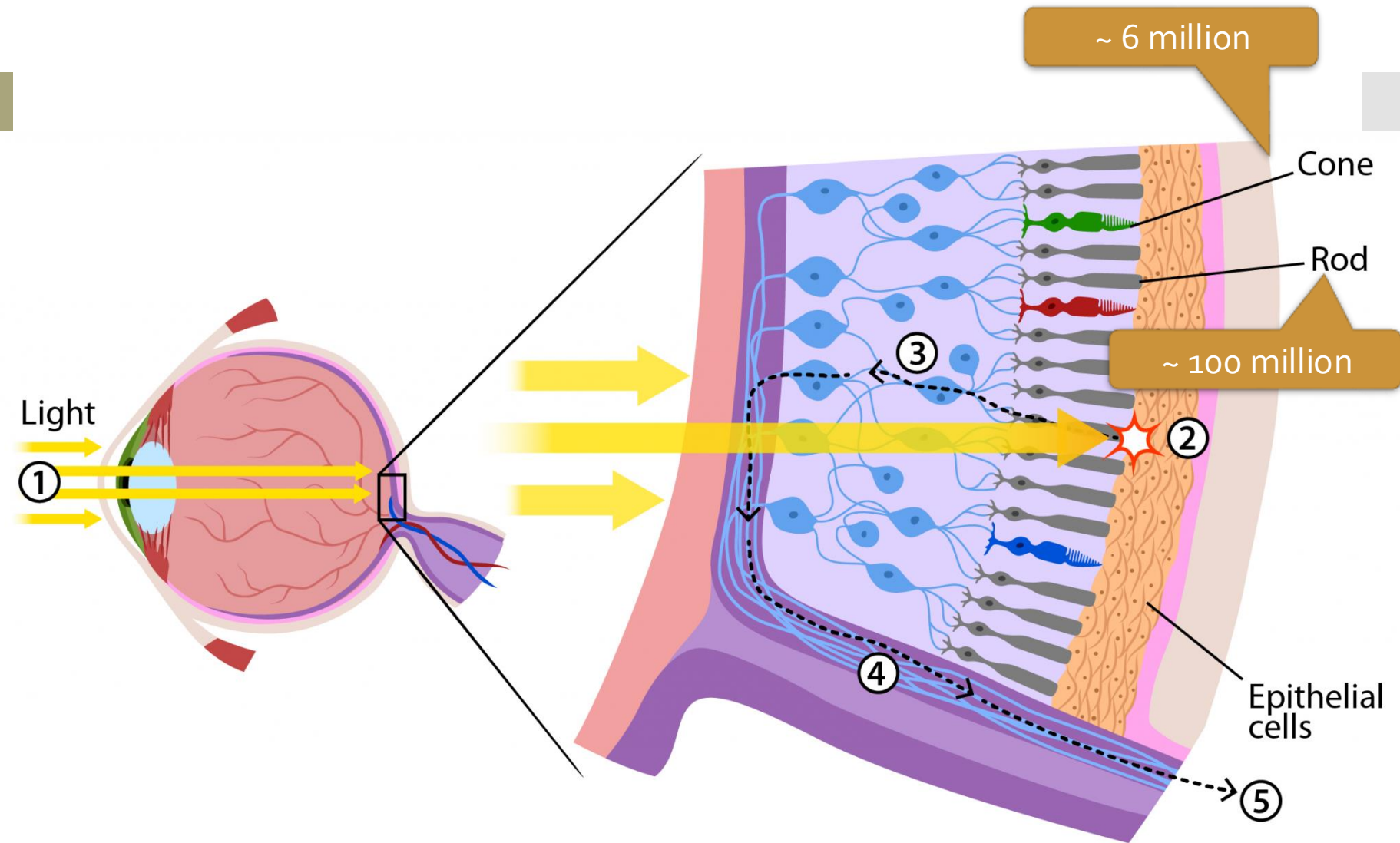
© Encyclopædia Britannica, Inc.

Visible Light



© Encyclopædia Britannica, Inc.

How we see color



Recall

What is expressiveness of a visual channel?

What is effectiveness of a visual channel?

Color for Visual Encoding

If we use color to represent our data, an important question to ask is: "How can we choose effective colors that provide good differentiation between data elements during the visualization task?"

C. G. Healey, "Choosing effective colours for data visualization," *Proceedings of Seventh Annual IEEE Visualization '96*, San Francisco, CA, USA, 1996, pp. 263-270, doi: 10.1109/VISUAL.1996.568118.

Color for Visual Encoding

If we use color to represent our data, an important question to ask is: “How can we choose effective colors that provide good differentiation between data elements during the visualization task?”

- How can we allow rapid and accurate identification of individual data elements through the use of color?
- What factors determine whether a “target” element’s color will make it easy to find, relative to differently colored “nontarget” elements?
- How many colors can we display at once, while still allowing for rapid and accurate target identification?

C. G. Healey, "Choosing effective colours for data visualization," *Proceedings of Seventh Annual IEEE Visualization '96*, San Francisco, CA, USA, 1996, pp. 263-270, doi: 10.1109/VISUAL.1996.568118.

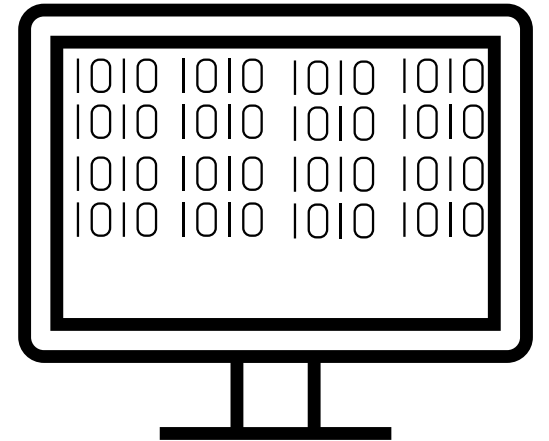
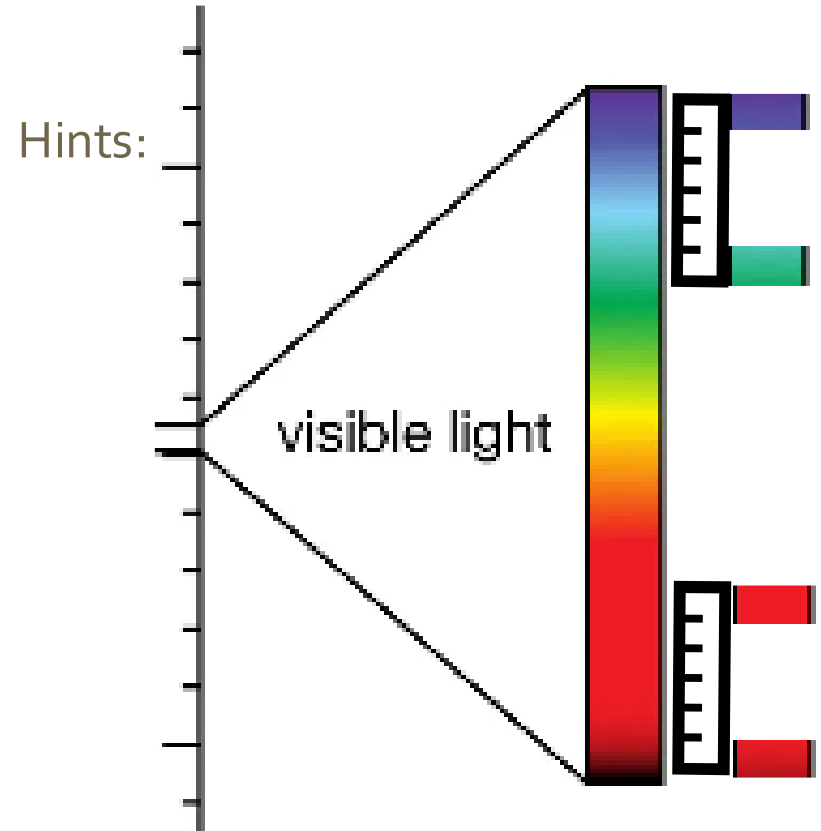
Colormaps

- Colormaps = predetermined sequences of color for representing data values
- 3 Main Principles:
 - **Order:** a good colormap sequence should be naturally orderable (e.g., from a cool blue color to warm red)
 - **Continuity:** the colormap should only reflect actual differences in the data without creating artificial boundaries
 - **Perceptual Uniformity:** adjacent colors should reflect even perceptual distances throughout the sequence, such that a step in data magnitude is matched by an equivalent perceptual step in color.

Colormaps

Perceptual Uniformity: adjacent colors should reflect even perceptual distances throughout the sequence, such that a step in data magnitude is matched by an equivalent perceptual step in color.

Why is this challenging?



Color for Visual Encoding

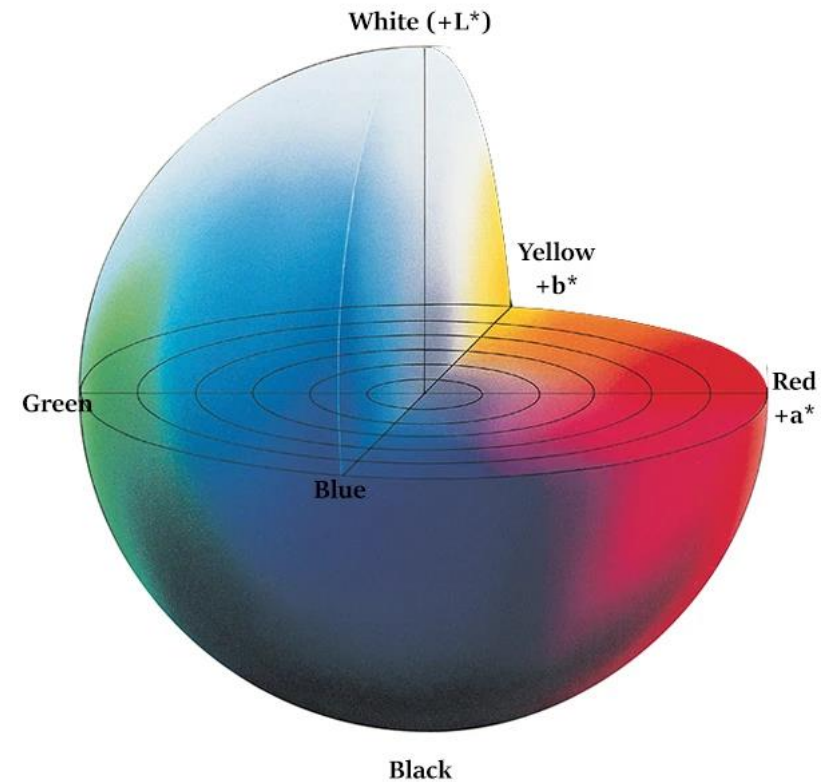
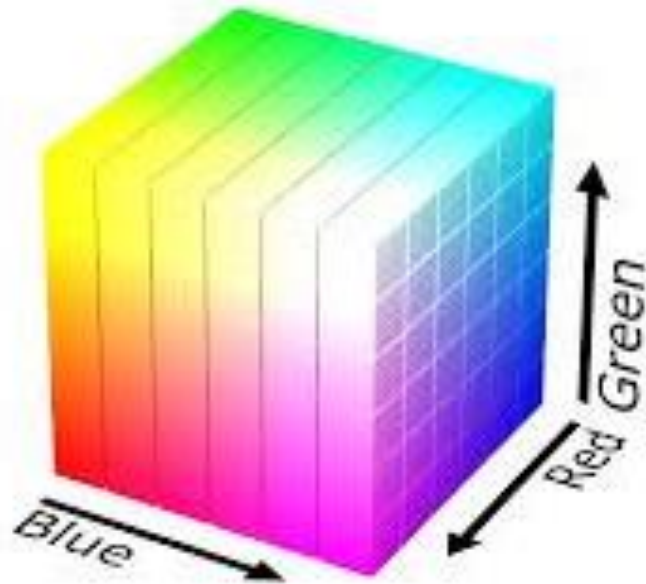
One approach—Control:

1. Color Distance
2. Linear separation of colors
3. Color category similarity

C. G. Healey, "Choosing effective colours for data visualization," *Proceedings of Seventh Annual IEEE Visualization '96*, San Francisco, CA, USA, 1996, pp. 263-270, doi: 10.1109/VISUAL.1996.568118.

Color for Visual Encoding

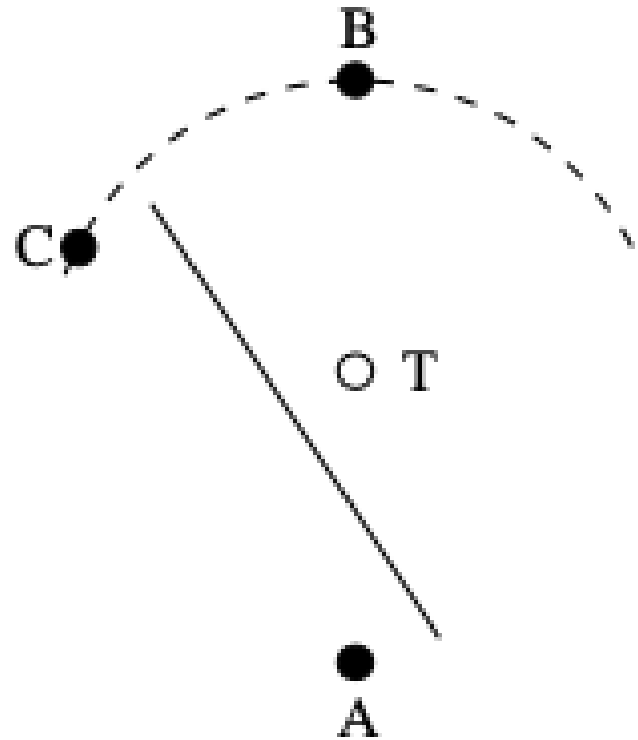
Color Distance: the Euclidean distance between different colors as measured in a perceptually balanced color model



C. G. Healey, "Choosing effective colours for data visualization," *Proceedings of Seventh Annual IEEE Visualization '96*, San Francisco, CA, USA, 1996, pp. 263-270, doi: 10.1109/VISUAL.1996.568118.

Color for Visual Encoding

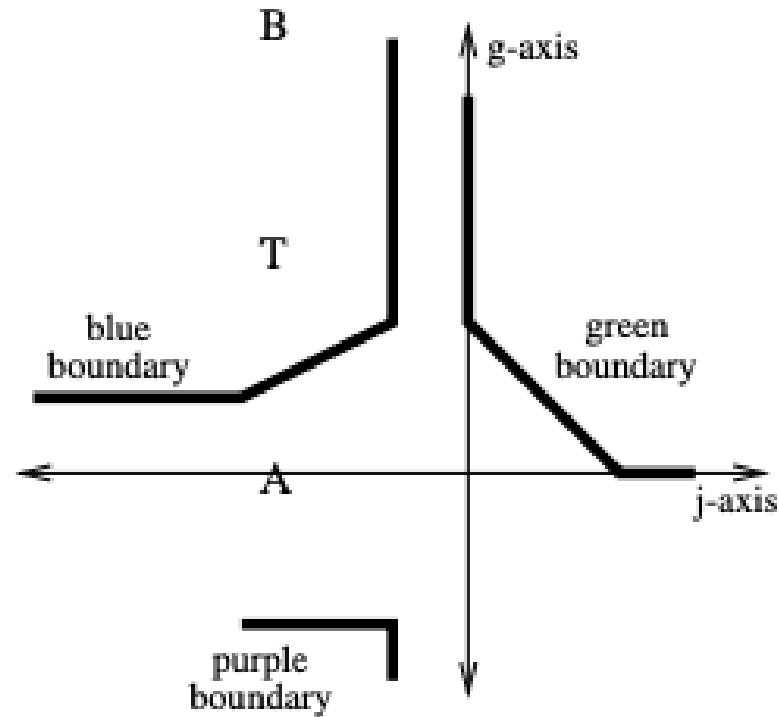
Linear Separation: the ability to linearly separate targets from non-targets in the color model being used



C. G. Healey, "Choosing effective colours for data visualization," *Proceedings of Seventh Annual IEEE Visualization '96*, San Francisco, CA, USA, 1996, pp. 263-270, doi: 10.1109/VISUAL.1996.568118.

Color for Visual Encoding

Color Category: the named color regions occupied by both the target and non-target elements



C. G. Healey, "Choosing effective colours for data visualization," *Proceedings of Seventh Annual IEEE Visualization '96*, San Francisco, CA, USA, 1996, pp. 263-270, doi: 10.1109/VISUAL.1996.568118.

Color for Visual Encoding

Another approach—Control:

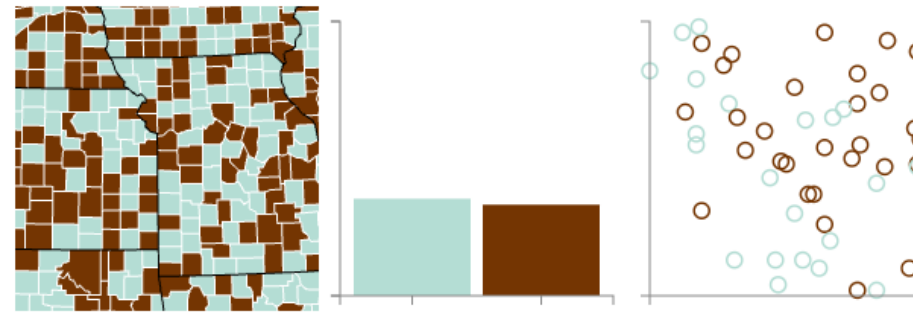
1. Perceptual Distance (CIEDE2000)
2. Name Difference
3. Pair Preference

The idea behind this approach is to prioritize aesthetically pleasing colormaps

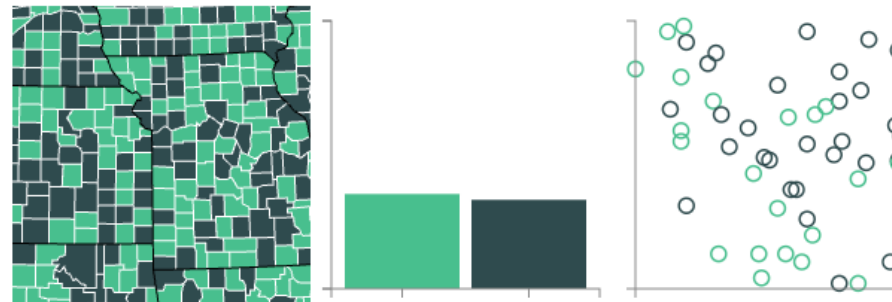
C. C. Gramazio, D. H. Laidlaw and K. B. Schloss, "Colorgorical: Creating discriminable and preferable color palettes for information visualization," in *IEEE Transactions on Visualization and Computer Graphics*, vol. 23, no. 1, pp. 521-530, Jan. 2017, doi: 10.1109/TVCG.2016.2598918.

Color for Visual Encoding

Perceptual Distance: A measure of how perceptually distinct two colors are. They use one based off CIEDE2000



High Perceptual Distance

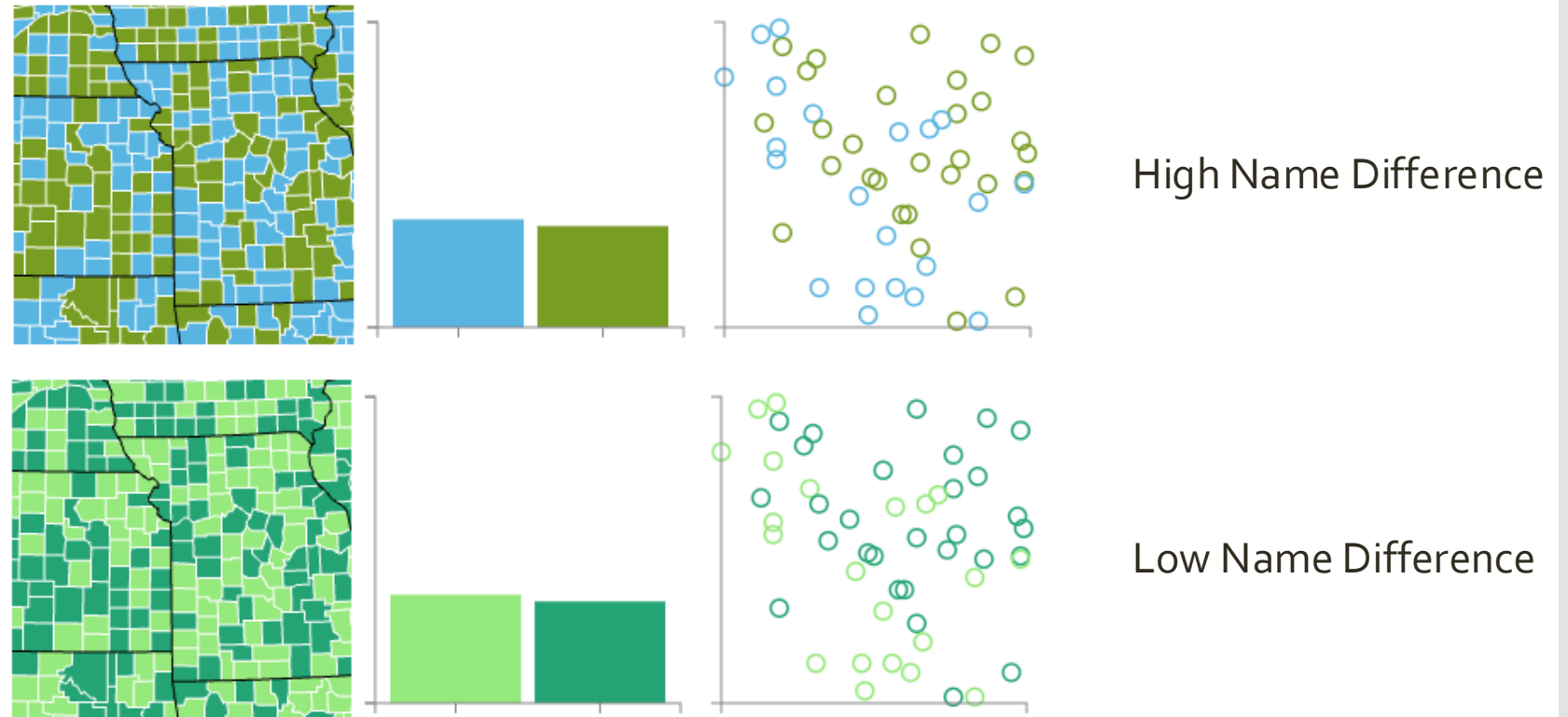


Low Perceptual Distance

C. C. Gramazio, D. H. Laidlaw and K. B. Schloss, "Colorgorical: Creating discriminable and preferable color palettes for information visualization," in *IEEE Transactions on Visualization and Computer Graphics*, vol. 23, no. 1, pp. 521-530, Jan. 2017, doi: 10.1109/TVCG.2016.2598918.

Color for Visual Encoding

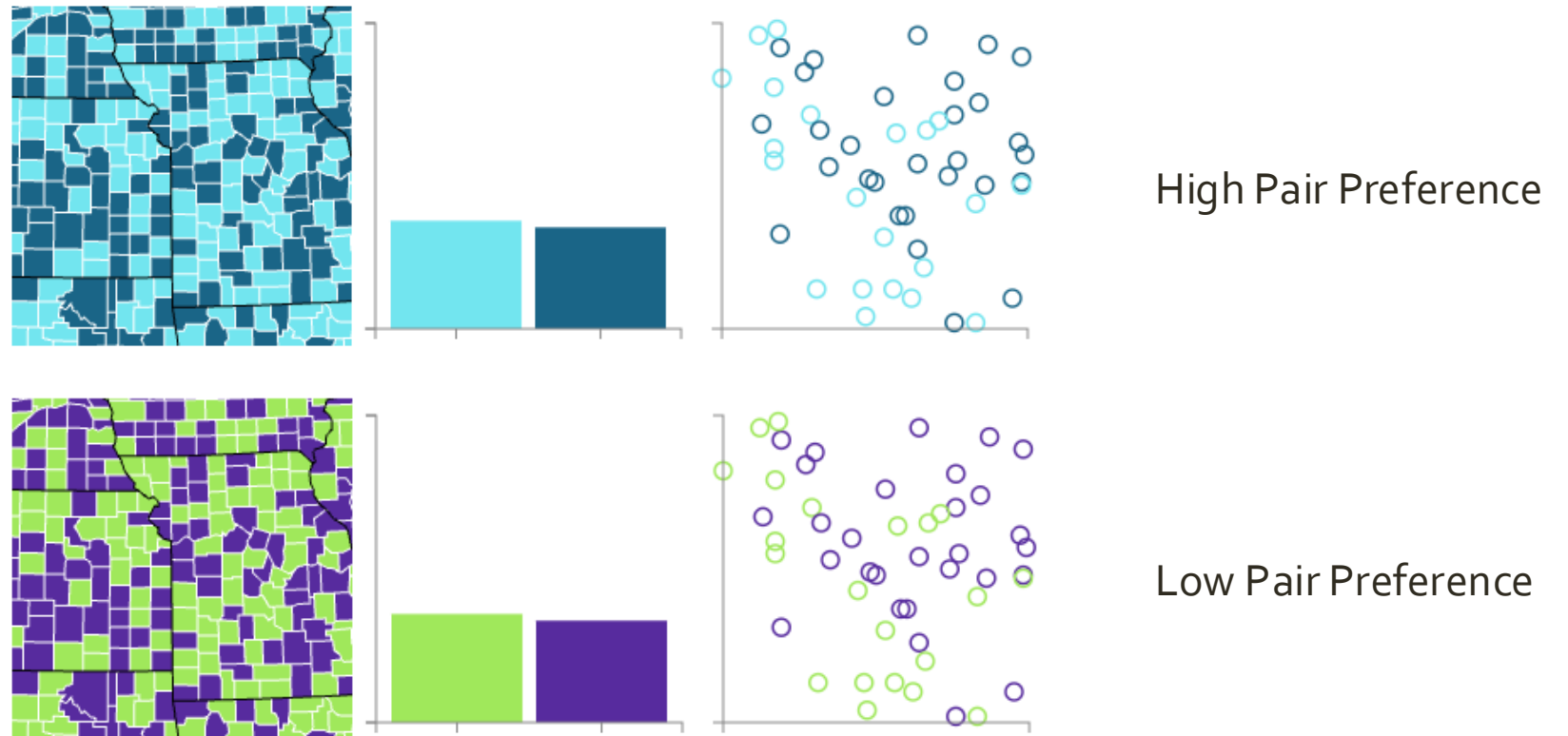
Name Difference: captures the degree to which two colors have distinct color-name association frequency distributions



C. C. Gramazio, D. H. Laidlaw and K. B. Schloss, "Colorgorical: Creating discriminable and preferable color palettes for information visualization," in *IEEE Transactions on Visualization and Computer Graphics*, vol. 23, no. 1, pp. 521-530, Jan. 2017, doi: 10.1109/TVCG.2016.2598918.

Color for Visual Encoding

Pair Preference: reflects people's preference for color combinations that contain cool colors that differ in lightness and are similar in hue



C. C. Gramazio, D. H. Laidlaw and K. B. Schloss, "Colorgorical: Creating discriminable and preferable color palettes for information visualization," in *IEEE Transactions on Visualization and Computer Graphics*, vol. 23, no. 1, pp. 521-530, Jan. 2017, doi: 10.1109/TVCG.2016.2598918.

Color for Visual Encoding

Another approach—Control:

1. Perceptual Distance (CIEDE2000)
2. Name Difference
3. Pair Preference

Try for yourself: <http://vrl.cs.brown.edu/color>

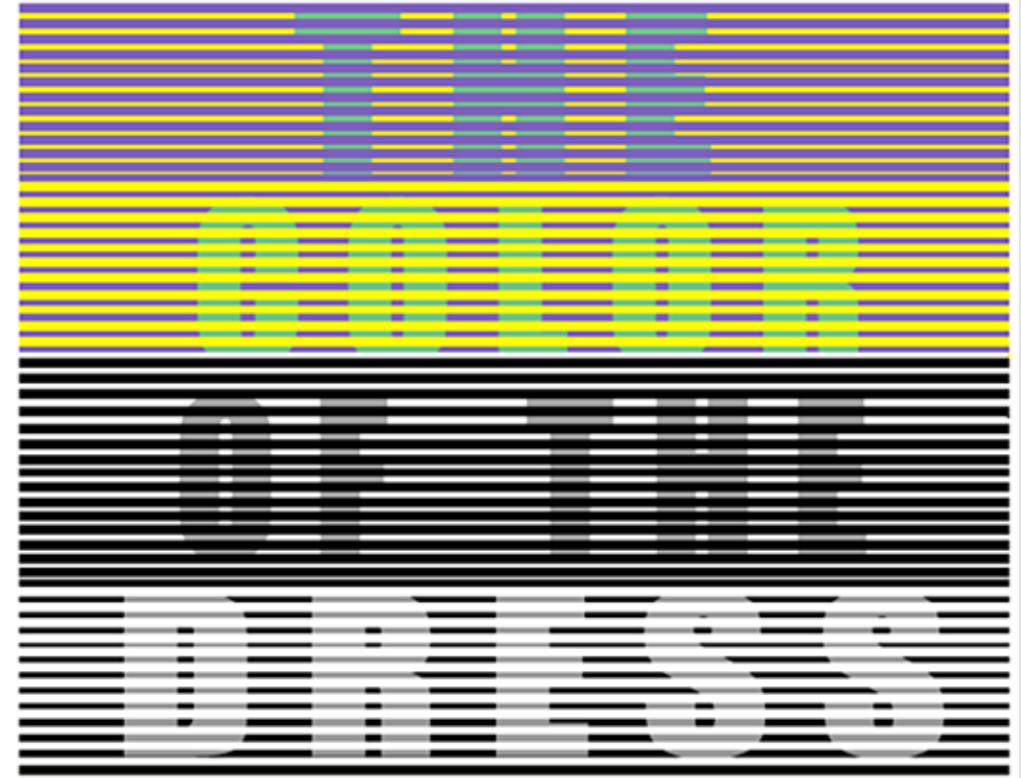
What do you notice?

C. C. Gramazio, D. H. Laidlaw and K. B. Schloss, "Colorgorical: Creating discriminable and preferable color palettes for information visualization," in *IEEE Transactions on Visualization and Computer Graphics*, vol. 23, no. 1, pp. 521-530, Jan. 2017, doi: 10.1109/TVCG.2016.2598918.

Color for Visual Encoding

Which approach do you think is preferable / more effective? Why?

Color phenomena



Color phenomena

Caveat 1: color is perceived in context



Which small square is **darker green**?

R.B. Lotto, & D. Purves, An empirical explanation of color contrast, Proc. Natl. Acad. Sci. U.S.A. 97 (23) 12834-12839, <https://doi.org/10.1073/pnas.210369597> (2000).

Color phenomena

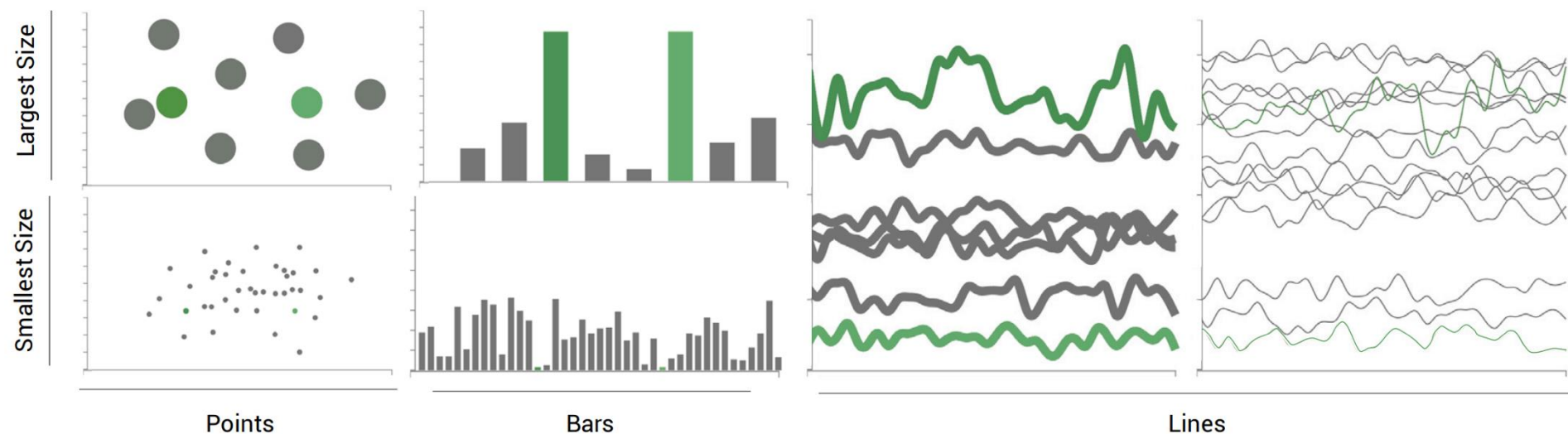
Caveat 2: difference is relative



R.B. Lotto, & D. Purves, An empirical explanation of color contrast, Proc. Natl. Acad. Sci. U.S.A. 97 (23) 12834-12839, <https://doi.org/10.1073/pnas.210369597> (2000).

Color phenomena

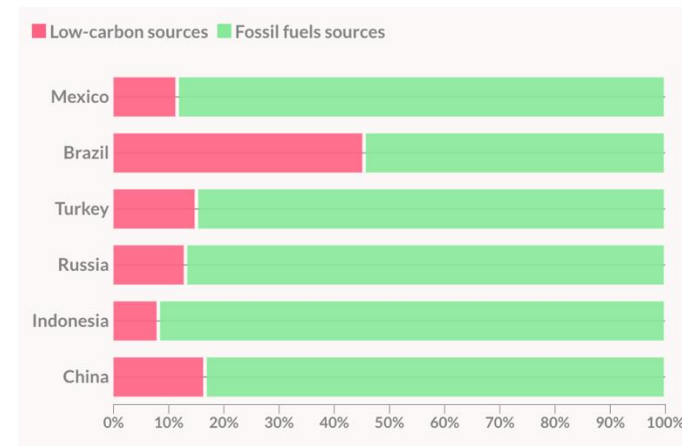
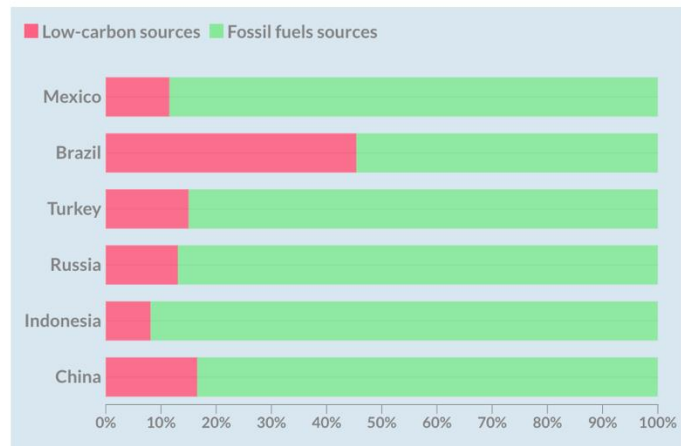
Caveat 3: Size of mark matters



D. A. Szafir, "Modeling Color Difference for Visualization Design," in *IEEE Transactions on Visualization and Computer Graphics*, vol. 24, no. 1, pp. 392-401, Jan. 2018, doi: 10.1109/TVCG.2017.2744359

- Be careful when layering colors

Color phenomenon



❌ (large text only)



CONTRAST
L^c 50.5

✅ (for icons & graphics)



CONTRAST
L^c 38.5

❌



CONTRAST
L^c 0.0

✅



CONTRAST
L^c 62.2

✅ (for icons & graphics)



CONTRAST
L^c 50.3

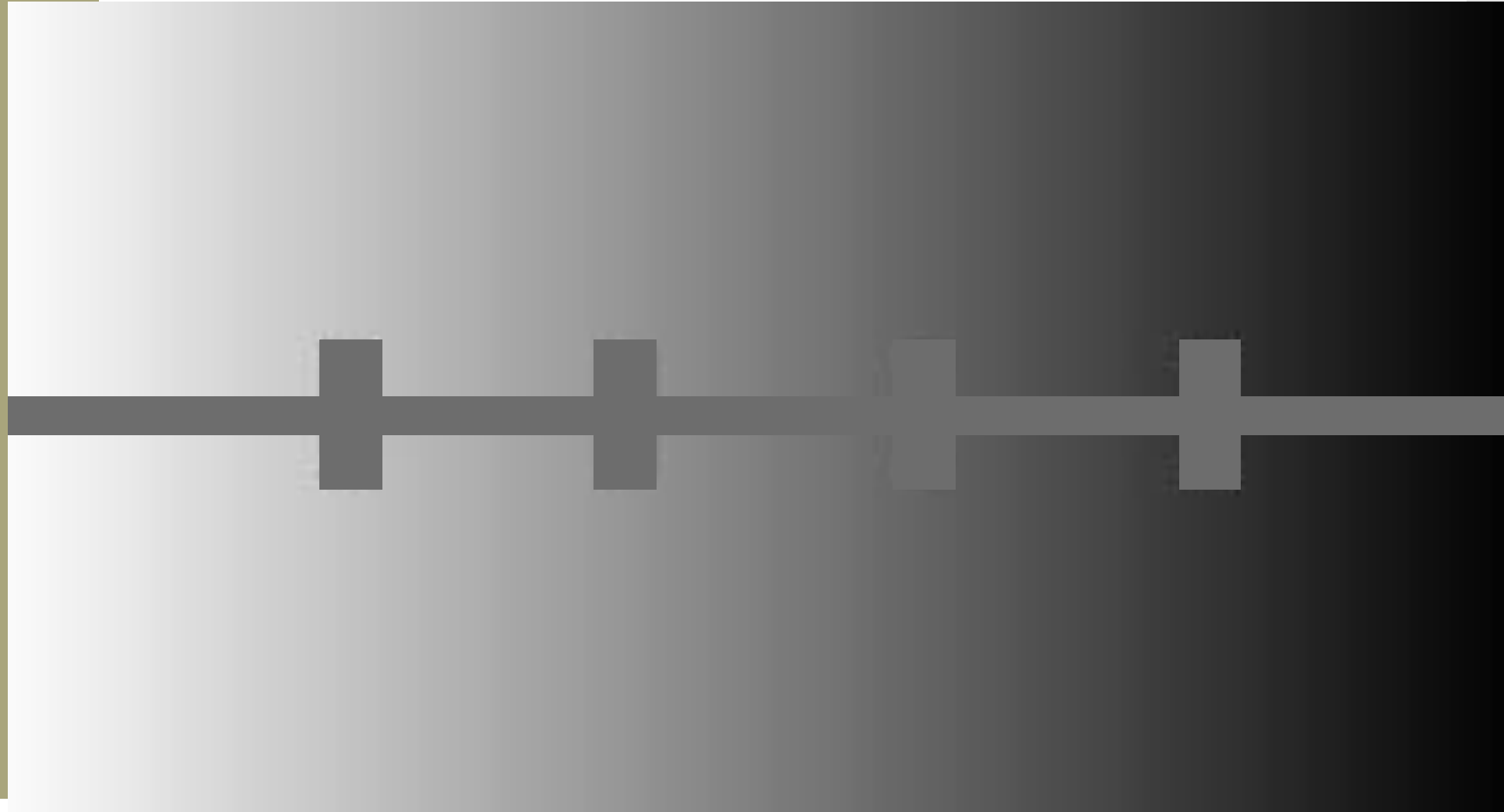
✅ (for icons & graphics)



CONTRAST
L^c 18.8

Caveat 2a: so are brightness and contrast

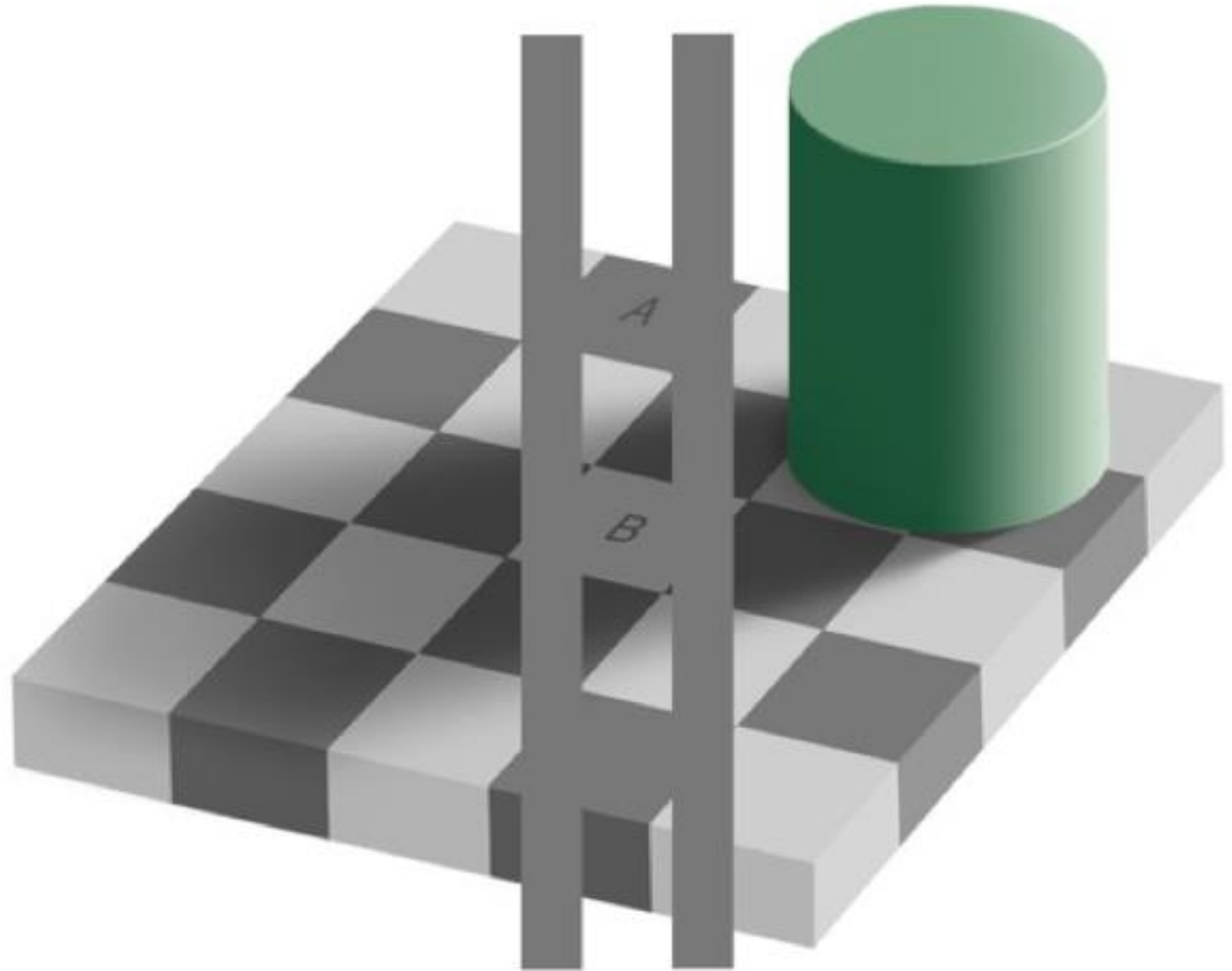
Color
phenomena



Ware, Colin. (1988). Color sequences for univariate maps: theory, experiments and principles. IEEE Computer Graphics and Applications. 8. 41-49. [10.1109/38.7760](https://doi.org/10.1109/38.7760).

Caveat 3: mental models > perception

Color
phenomena



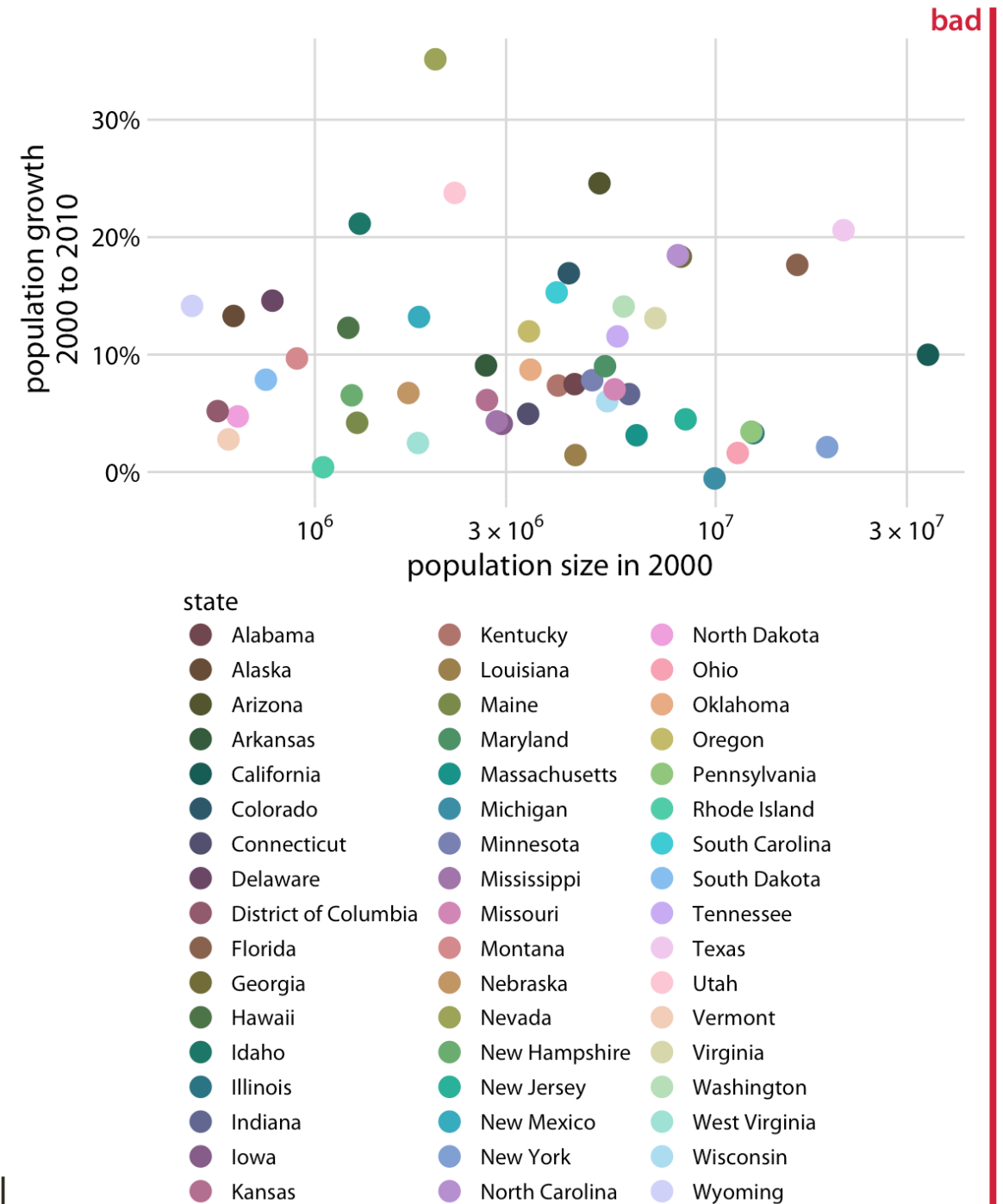
Adelson, Edward H. 1995. "Checkers shadow Illusion."

Color Problems

- Using a poor color scheme can also cause issues with your visualization

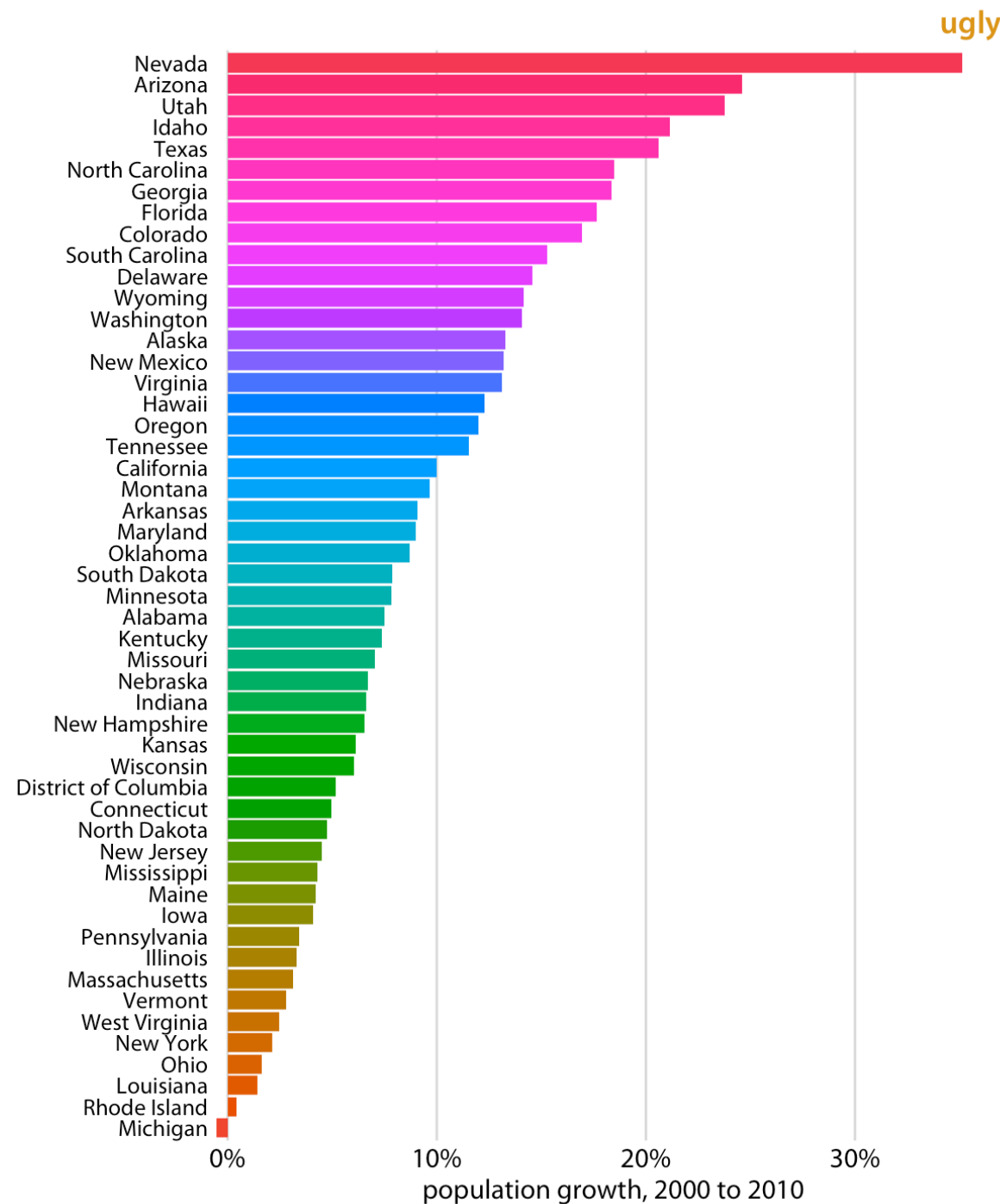
Color Problems

- Hue mapped to too many categories



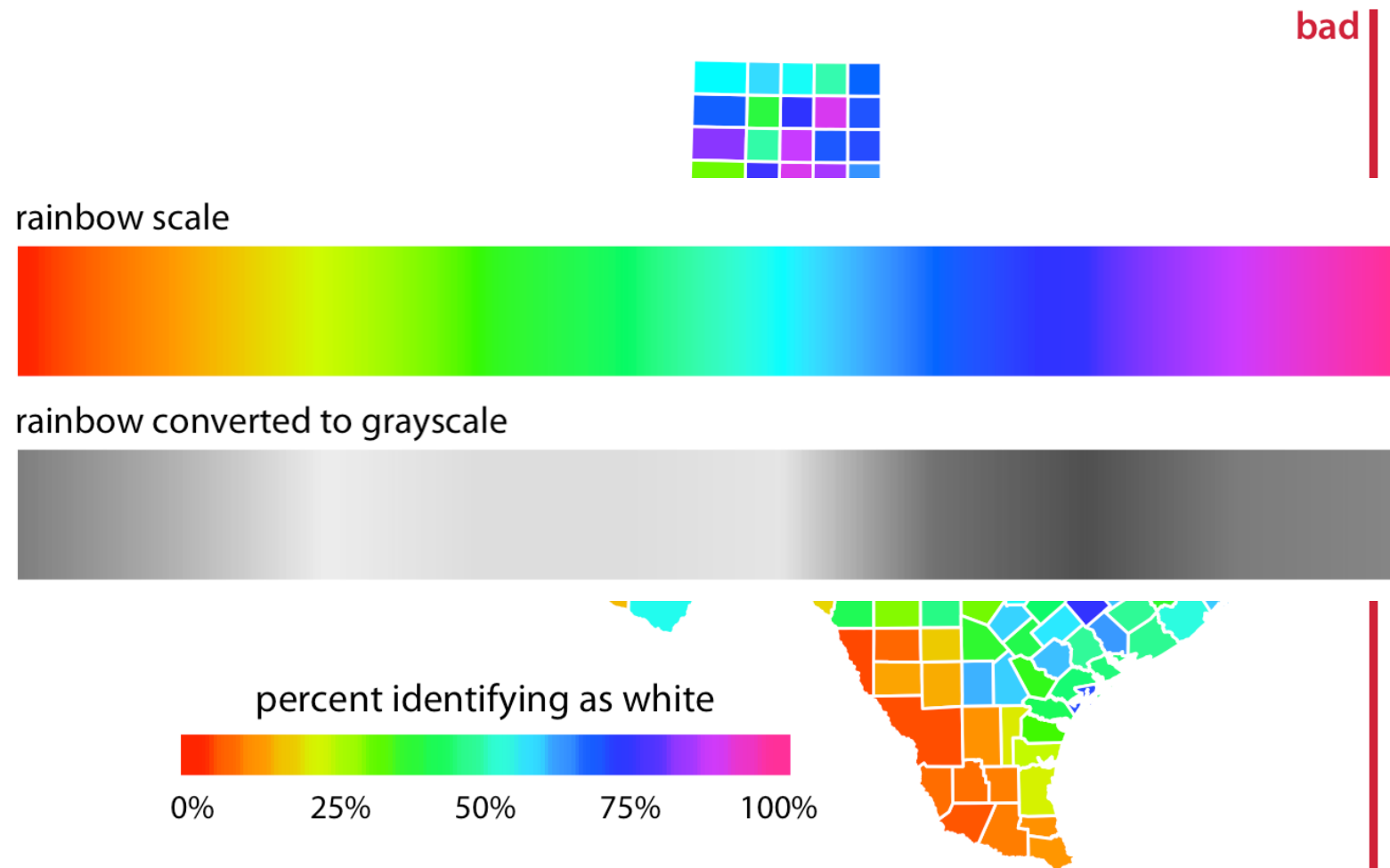
Color Problems

- Hue not mapped to data



Color Problems

- Rainbow obscures real patterns (it is not monotonic)



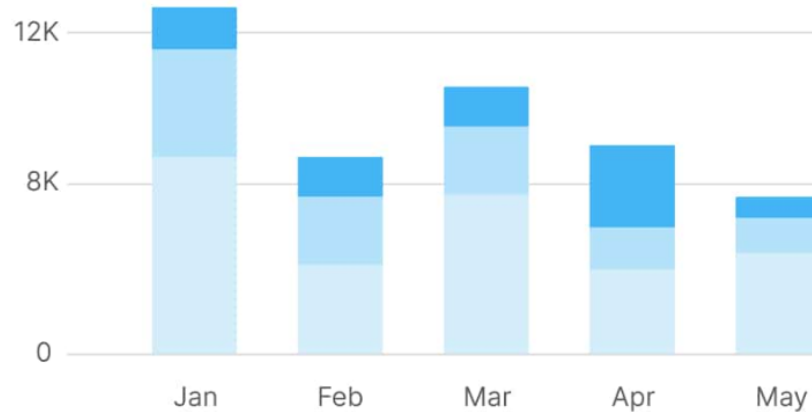
Color Problems

- Similar colors are hard to differentiate

✕ Similar colors

Product Sales

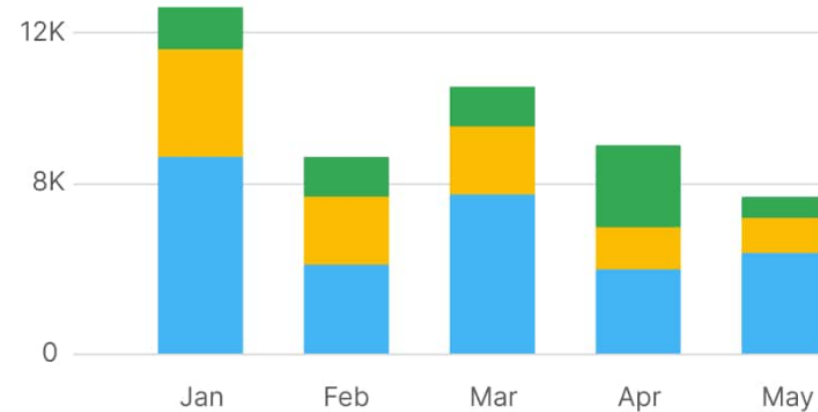
□ Clothing □ Equipment ■ Accessories



✓ Different colors

Product Sales

■ Clothing ■ Equipment ■ Accessories



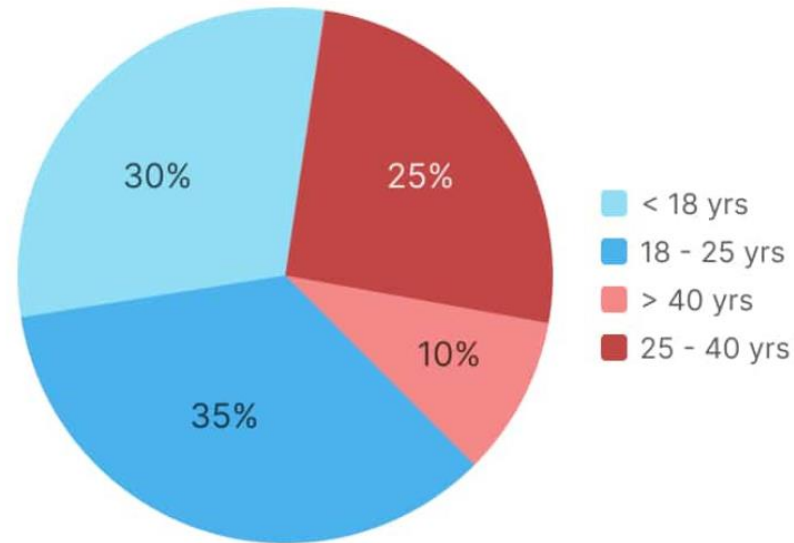
<https://matthewstrom.com/writing/how-to-pick-the-least-wrong-colors/>

Color Problems

- Similar colors imply relationships

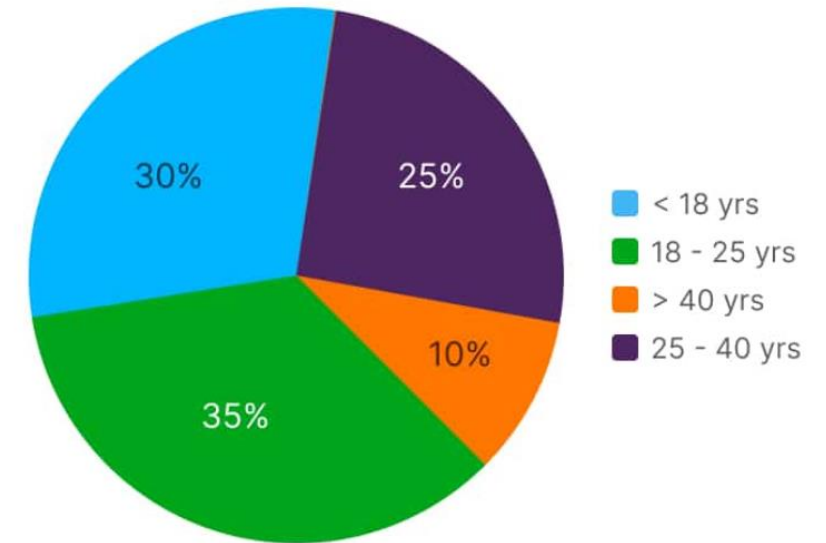
✗ Uneven color distribution

Age distribution of users

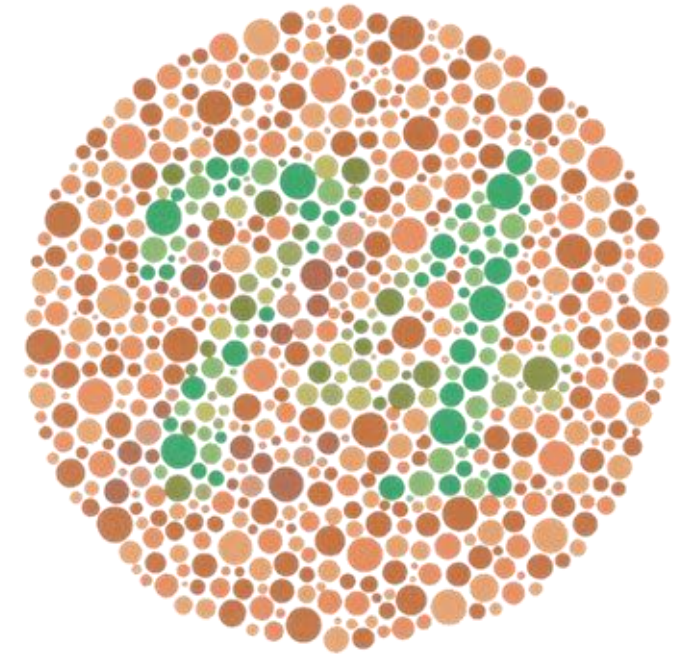


✓ Even color distribution

Age distribution of users



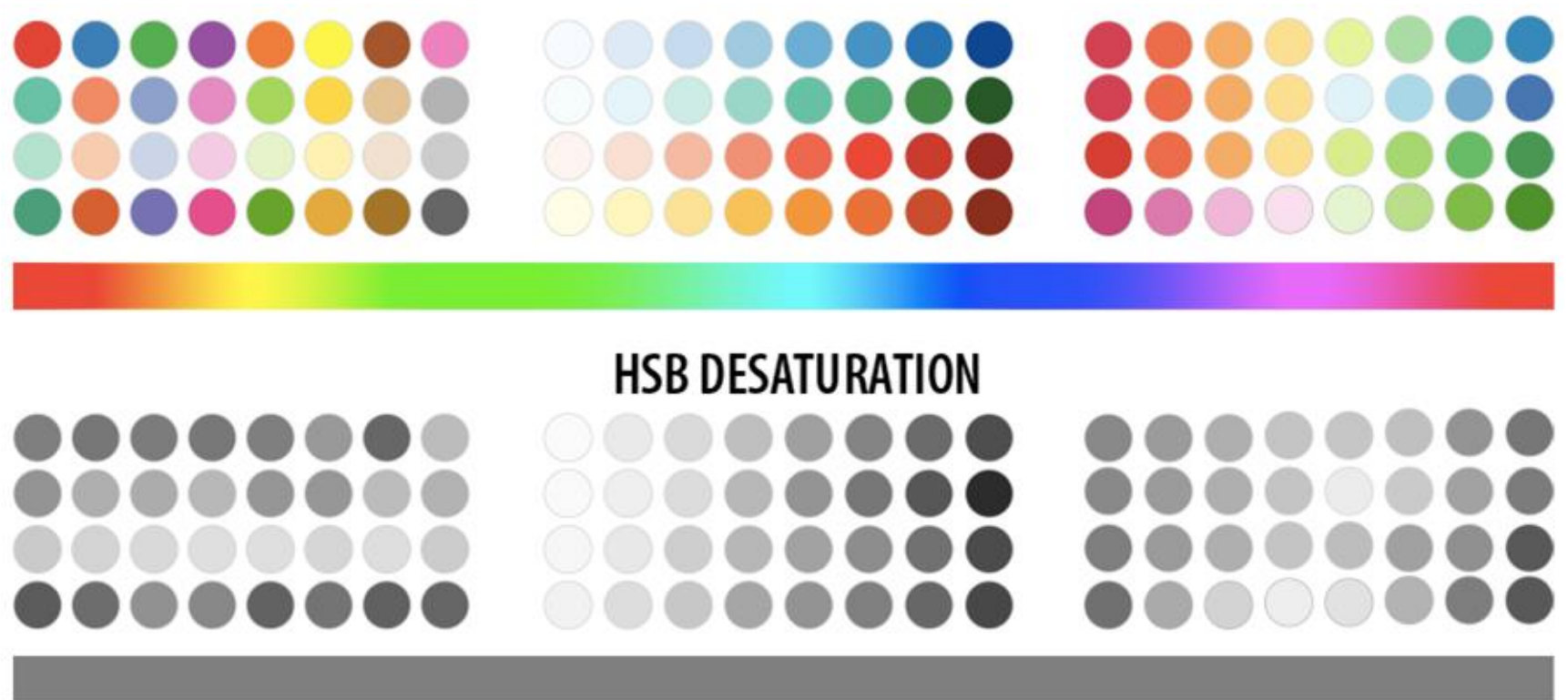
“colorblindness”



1 out of every 8 people has just 2 types of color receptors (rather than 3)

What happens
when you
print?

- Need color scheme that converts well to grey scale



Colorbrewer palettes

- colorbrewer.org provides a whole bunch of palettes that can help us avoid these issues
- This makes life a lot easier for us!

QUALITATIVE

set1



set2



pastel2



dark2



SEQUENTIAL

blues



greens



reds



ylorbr



DIVERGING

spectral



rdylbu



rdylgn



piyg



Colorbrewer palettes

QUALITATIVE

set1



set2



pastel2



dark2

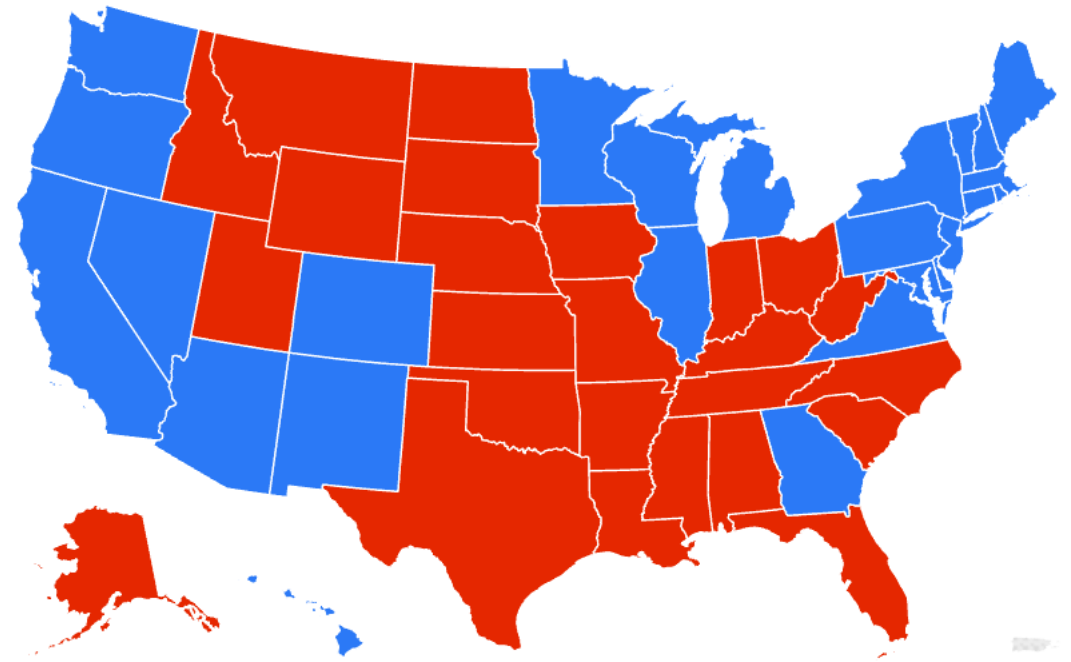


Good for nominal data

Categorical color palette

2020 Presidential election: winner per state

■ Democrat ■ Republican



Source: [The Cook Political Report](#) • Created with the Flourish [Projection Map](#) template

Colorbrewer palettes

SEQUENTIAL

blues



greens



reds



ylorbr

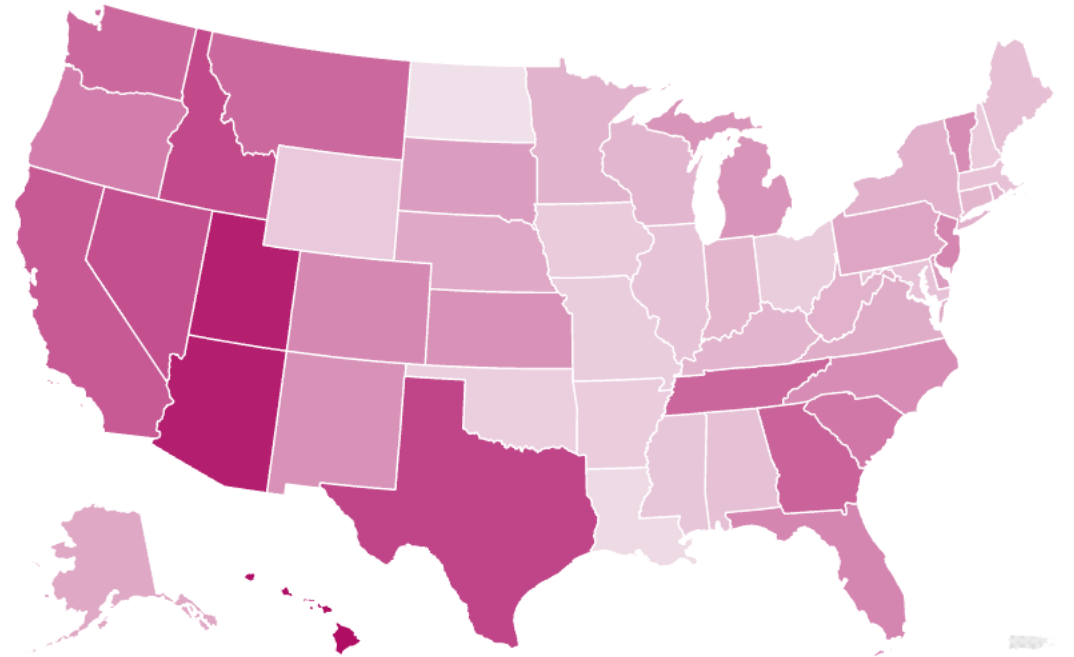


Good for sequential ordinal
& quantitative data

Sequential color palette

2020 Presidential election: vote change (%)

5.1  33.9



Source: [The Cook Political Report](#) • Created with the Flourish [Projection Map](#) template

Colorbrewer palettes

DIVERGING

spectral



rdylbu



rdylgn



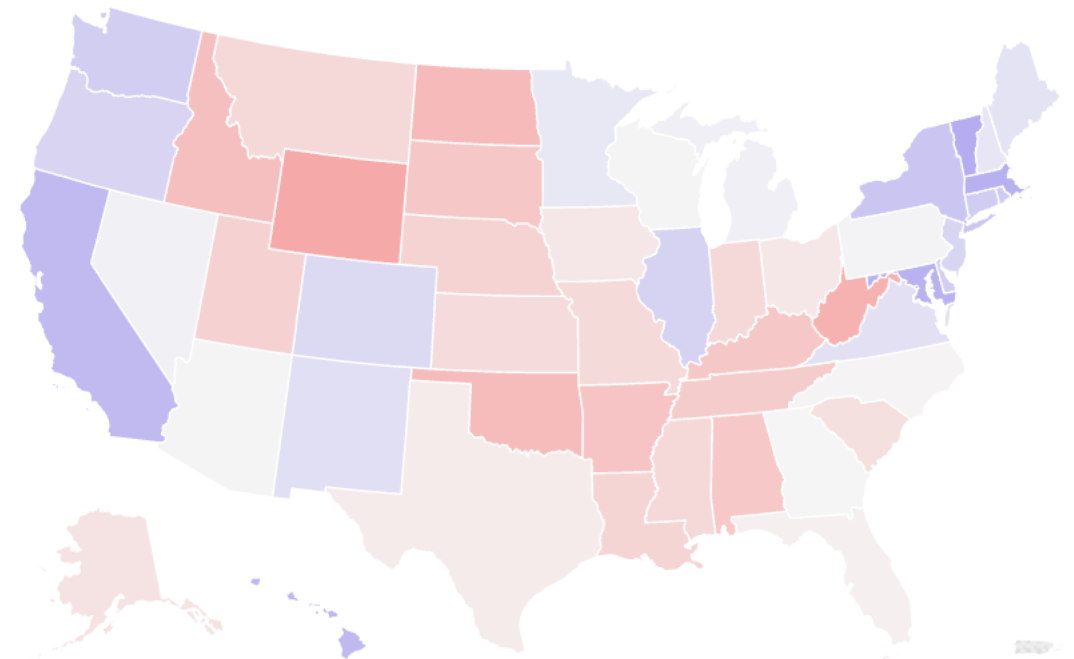
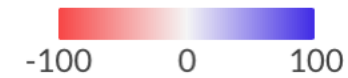
piyg



Good for diverging ordinal
& quantitative data

Diverging color palette

2020 Presidential election: margin of victory (%)



Source: [The Cook Political Report](#) • Created with the Flourish [Projection Map](#) template

Takeaways: Perception

- Visualization is about more than just aesthetics
- There are compelling **cognitive reasons** why some visualization techniques are helpful and others aren't
- The choices we make about **visual mappings** can have a significant effect on performance

Your turn!

- Work with a classmate or 2
- Find a two visualizations from trusted sources that visually encode data via color. One should use hue and the other saturation.
- Assess the use of color:
 - Is it perceptually accurate and easy to read?
 - Is it aesthetically pleasing?
 - Are there any optical illusions?
 - What would you change if you were to re-design the visualization?